

$$k = \frac{24EI_c}{h^3} \frac{12\rho + 1}{12\rho + 4} \quad (1.3.5)$$

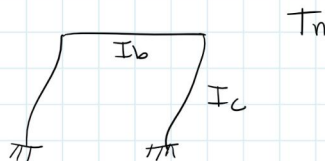
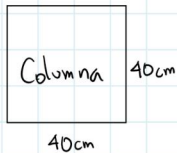
$$\rho = (EI_b/L) \div (2EI_c/h)$$

$$e = \left[\frac{(EI_c/2)}{2EI_c/h} \right] \div \left[\frac{2EI_c/h}{2EI_c/h} \right] = \frac{1/2}{2} = \frac{1}{4} = 1/8$$

$$K = \frac{24EI_c}{h^3} \left(\frac{12 \cdot 1/8 + 1}{12 \cdot 1/8 + 4} \right) = \frac{24EI_c}{h^3} \left(\frac{3/2 + 1}{3/2 + 4} \right) = \frac{24EI_c}{h^3} \cdot \left(\frac{5/2}{11/2} \right)$$

$$K = \frac{24EI_c}{h^3} \cdot \frac{5}{11}$$

$$K = \frac{120}{11} \frac{EI_c}{h^3}$$



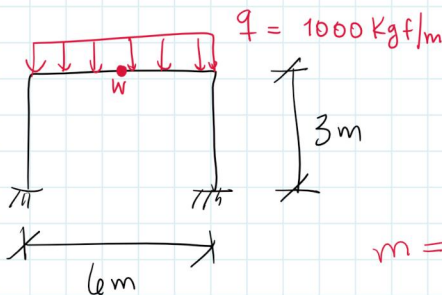
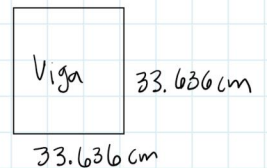
$$E = 19,100 \sqrt{281} = 253,122.125 \text{ Kg/cm}^2$$

$$I_c = \frac{1}{12} \cdot 40 \cdot 40^3 = 213,333.333 \text{ cm}^4 \quad \text{Columnas}$$

$$I_b = \frac{I_c}{2} = \frac{213,333.333}{2} = 106,666.666 \text{ cm}^4 \quad \text{Viga}$$

$$I_b = \frac{1}{12} b h^3 \quad b = h \quad = \frac{1}{12} h \cdot h^3 = \frac{h^4}{12} = 106,666.666 \text{ cm}^4$$

$$h = \sqrt[4]{12 \cdot 106,666.666} = 33.636 \text{ cm}$$



$$\omega_n = \sqrt{\frac{120EI_c}{11mh^3}}$$

$$T_n = \frac{2\pi}{\omega_n}$$

$$m = \frac{W}{g}$$

$$W = q \cdot L = 1000 \cdot 6 = 6000 \text{ Kg f}$$

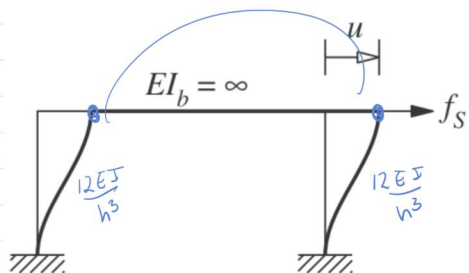
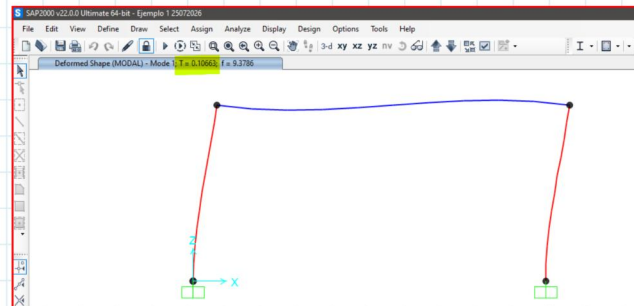
$$m = \frac{6000 \text{ Kg f}}{981 \text{ cm/s}^2}$$

$$\omega_n = \sqrt{\frac{120 \cdot 253,122.125 \cdot 213,333.333}{11 \cdot \frac{6000 \text{ Kg f}}{981 \text{ cm/s}^2} + 300^3 \text{ cm}^3 \cdot \frac{1}{52}}}$$

$$\omega_n = 59.726 \text{ s}^{-1}$$

$$T_n = \frac{2\pi}{59.726} = 0.1052 \text{ seg}$$

$$T_n = \frac{2\pi}{59.726} = 0.1052 \text{ seg}$$



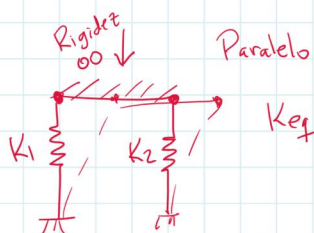
$$k = \frac{24EI_c}{h^3} \frac{12\rho + 1}{12\rho + 4}$$

$$\rho = (EI_b/L) \div (2EI_c/h)$$

$$\omega_n = \sqrt{\frac{24EI_c}{h^3} \frac{12\rho + 1}{12\rho + 4}} = \sqrt{\frac{24EI_c}{m h^3}}$$

$$k = \sum_{\text{columns}} \frac{12EI_c}{h^3} = 24 \frac{EI_c}{h^3}$$

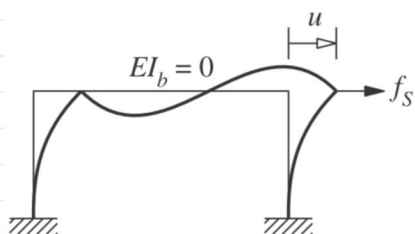
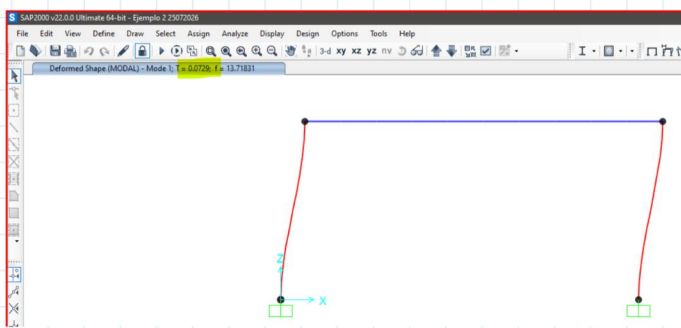
(1.3.2)



$$K_{eq} = K_1 + K_2 = \frac{12EI}{h^3} + \frac{12EI}{h^3} = \frac{24EI}{h^3}$$

$$\omega_n = \sqrt{\frac{24 \times 253,122.125 \times 213,333.333}{\frac{6000}{981} + 300^3}} = 88.588 \text{ s}^{-1}$$

$$T_n = \frac{2\pi}{88.588} = 0.0709 \text{ seg}$$



$$k = \frac{24EI_c}{h^3} \frac{12\rho + 1}{12\rho + 4}$$

$$\rho = (EI_b/L) \div (2EI_c/h)$$

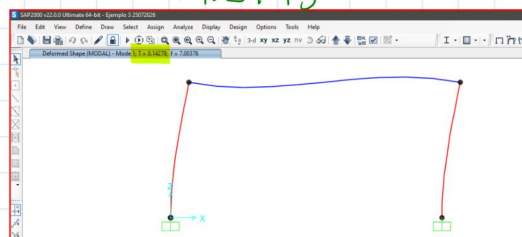
$$K = \frac{6EI_c}{h^3}$$

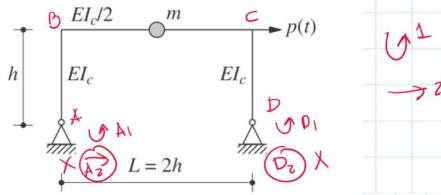
$$e = 0/L \div \frac{2EI_c}{h} = 0$$

$$\omega_n = \sqrt{\frac{6EI_c}{h^3} \frac{12\rho + 1}{12\rho + 4}} = \sqrt{\frac{6EI_c}{m h^3}}$$

$$\omega_n = \sqrt{\frac{6 \times 253,122.125 \times 213,333.333}{\frac{6000}{981} + 300^3}} = 44.294 \text{ s}^{-1}$$

$$T_n = \frac{2\pi}{44.294} = 0.142 \text{ seg}$$





Matriz global de la estructura

$B_2 = C_2$

	A1	A2	B1	B2	C1	C2	D1	D2	
A1	$4h^2$	$-6h$	$2h^2$	$6h$	0	0	0	0	
A2	$-6h$	12	$-6h$	-12	0	0	0	0	
B1	$2h^2$	$-6h$	$5h^2$	$6h$	$h^2/2$	0	0	0	
B2	$6h$	-12	$6h$	12	0	0	0	0	
C1	0	0	$h^2/2$	0	$5h^2$	$6h$	$2h^2$	$-6h$	
C2	0	0	0	0	$6h$	12	$6h$	-12	
D1	0	0	0	0	$2h^2$	$6h$	$4h^2$	$-6h$	
D2	0	0	0	0	$-6h$	-12	$-6h$	12	

$\frac{EI_c}{h^3}$

Matriz reducida

Eliminación de restricciones de apoyos

Suma B2 y C2
 $B_2 + C_2$

	A1	B1	B2	C1	C2	D1	
A1	$4h^2$	$2h^2$	$6h$	0	0	0	
B1	$2h^2$	$5h^2$	$6h$	$h^2/2$	0	0	
B2	$6h$	$6h$	12	0	0	0	
C1	0	$h^2/2$	0	$5h^2$	$6h$	$2h^2$	
C2	0	0	0	$6h$	12	$6h$	
D1	0	0	0	$2h^2$	$6h$	$4h^2$	

$\frac{EI_c}{h^3} \times$

	A1	B1	C1	D1	
A1	$6h$	$4h^2$	$2h^2$	0	0
B1	$6h$	$2h^2$	$5h^2$	$h^2/2$	0
B2	12	$6h$	$6h$	0	0
C1	$6h$	0	$h^2/2$	$5h^2$	$2h^2$
C2	12	0	0	$6h$	$6h$
D1	$6h$	0	0	$2h^2$	$4h^2$

Suma filas B2 y C2
 $B_2 + C_2$

$\frac{EI_c}{h^3} \times$

	B2+C2	A1	B1	C1	D1
B2+C2	24	$6h$	$6h$	$6h$	$6h$
A1	$6h$	$4h^2$	$2h^2$	0	0
B1	$6h$	$2h^2$	$5h^2$	$h^2/2$	0
C1	$6h$	0	$h^2/2$	$5h^2$	$2h^2$
D1	$6h$	0	0	$2h^2$	$4h^2$

$\frac{6EI_c}{h^3} *$

	B_2+C_2	A_1	B_1	C_1	D_1
$B_2+C_2 \rightarrow$	24	6h	6h	6h	6h
$A_1 \rightarrow$	6h	4h ²	2h ²	0	0
$B_1 \rightarrow$	6h	2h ²	5h ²	h ² /2	0
$C_1 \rightarrow$	6h	0	h ² /2	5h ²	2h ²
$D_1 \rightarrow$	6h	0	0	2h ²	4h ²

$\begin{bmatrix} a & b \\ K_{aa} & K_{ab} \\ K_{ba} & K_{bb} \end{bmatrix}$

$$K_{aa} = [24] \quad K_{ab} = [6h, 6h, 6h, 6h]$$

$$K_{ba} = \begin{bmatrix} 6h \\ 6h \\ 6h \\ 6h \end{bmatrix} \quad K_{bb} = \begin{bmatrix} 4h^2 & 2h^2 & 0 & 0 \\ 2h^2 & 5h^2 & h^2/2 & 0 \\ 0 & h^2/2 & 5h^2 & 2h^2 \\ 0 & 0 & 2h^2 & 4h^2 \end{bmatrix}$$

$$K_c = K_{aa} - K_{ab} K_{bb}^{-1} K_{ba}$$

$$K_c = \frac{EI_c}{h^3} \left[24 - [6h \ 6h \ 6h \ 6h] * \begin{bmatrix} 79 & 8 & 1 & 1 \\ 252h^2 & 63h^2 & 63h^2 & 126h^2 \\ 8 & 16 & 2 & 1 \\ 63h^2 & 63h^2 & 63h^2 & 63h^2 \\ 1 & 2 & 16 & 8 \\ 63h^2 & 63h^2 & 63h^2 & 63h^2 \\ 1 & 1 & 8 & 79 \\ 126h^2 & 63h^2 & 63h^2 & 252h^2 \end{bmatrix} * \begin{bmatrix} 6h \\ 6h \\ 6h \\ 6h \end{bmatrix} \right]$$

$$K_c = \frac{EI_c}{h^3} [24 - 22] = \frac{2EI_c}{h^3}$$

$$K_c = \frac{2EI_c}{h^3}$$

$$\omega_n = \sqrt{\frac{2EI_c}{m h^3}} \quad T_n = \frac{2\pi}{\omega_n}$$

$$\omega_n = \sqrt{\frac{2 * 253,122.125 * 213,333.333}{\frac{6000}{981} * 300^3}} = 25.573 \text{ s}^{-1}$$

$$T_n = \frac{2\pi}{25.573} = 0.246 \text{ seg}$$

